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TEMPERATURE MEASURING CIRCUIT

Abstract

A temperature measuring circuit includes: a first switch circuit configured to charge a capacitor upon receiving a first control signal and discharge the capacitor upon receiving a second control signal; a control circuit configured to set a control signal at a second control signal in response to an output voltage indicating that an inter-terminal voltage is higher than a first reference voltage, set the control signal at the first control signal in response to the output voltage indicating that the inter-terminal voltage is lower than a second reference voltage, and set the control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than a third reference voltage; a counter circuit configured to count a frequency of the control signal; and a temperature obtaining unit configured to obtain the temperature based on the frequency.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] The present application claims priority from Japanese Application JP 2024-029515, the content of which is hereby incorporated by reference into this application.
BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present disclosure relates to a temperature measuring circuit.
2. Description of the Related Art
[0003] United States Unexamined Patent Application Publication No. 2022/0163402 discloses a temperature sensor.
[0004] In the temperature sensor, a comparator outputs a first readout number when the voltage at a first input terminal is lower than the voltage at the second input terminal, and the comparator outputs a second readout number when the voltage at the second input terminal is lower than the voltage at the first input terminal. When the comparator outputs the first readout number, a read circuit is in an A-phase state. When the comparator outputs the second readout number, the read circuit is in a B-phase state.
[0005] When the readout circuit is in the A-phase state, a proportional current IPTAT is transmitted to a first end of a capacitor. Further, a second end of the capacitor is connected to a ground. Further, a voltage, which is the potential difference from the first to second ends of the capacitor, is input to the first input terminal of the comparator. Further, the voltage between the base and emitter of a bipolar junction transistor is input to the second input terminal of the comparator.
[0006] When the readout circuit is in the B-phase state, the proportional current IPTAT is transmitted to the second end of the capacitor. Further, the first end of the capacitor is connected to the ground. Further, ground voltage is input to the first input terminal of the comparator. Further, a voltage, which is the potential difference from the second to first ends of the capacitor, is input to the second input terminal of the comparator.
[0007] A controller determines a temperature based on the oscillation period of the readout number output by the comparator (paragraphs [0036], [0046], [0047], [0052], [0057], [0058], [0059], [0062], and [0063], as well as FIGS. **4** and **5**).
SUMMARY OF THE INVENTION
[0008] In the temperature sensor disclosed in United States Unexamined Patent Application Publication No. 2022/0163402, the capacitor's second end receives a negative voltage when the readout circuit is in the B-phase state. This makes it difficult to form the temperature sensor onto a semiconductor substrate.
[0009] Further, in the temperature sensor, the proportional current IPTAT, which is transmitted to the capacitor's second end when the readout circuit is in the B-phase state, depends on the temperature property of a resistor provided in the circuit that generates the proportional current IPTAT. This makes it impossible to determine the temperature with high accuracy.
[0010] One aspect of the present disclosure has been made in view of this problem. The aspect of the present disclosure aims to provide a temperature measuring circuit that can be easily formed on a semiconductor substrate, and that can measure a temperature with high accuracy.
[0011] A temperature measuring circuit according to one aspect of the present disclosure is provided with the following: a capacitor configured to generate an inter-terminal voltage corresponding to the amount of stored electric charge; a first current supply configured to feed a first current; a second current supply configured to feed a second current; a first voltage supply configured to generate a first reference voltage; a second voltage supply configured to generate a second reference voltage; a third voltage supply configured to generate a third reference voltage

corresponding to a temperature; and a comparator circuit configured to output an output voltage indicating the levels of two input voltages that are received. The temperature measuring circuit is also provided with a first switch circuit configured to, upon receiving a first control signal, charge the capacitor through the first current and set the two input voltages at a selected reference voltage and the inter-terminal voltage, and upon receiving a second control signal, discharge the capacitor through the second current and set the two input voltages at the second reference voltage and the inter-terminal voltage. The temperature measuring circuit is also provided with a second switch circuit configured to switch between a first connection state in which the selected reference voltage is set at the first reference voltage, and a second connection state in which the selected reference voltage is set at the third reference voltage. The temperature measuring circuit is also provided with a control circuit configured to, during a first period, set a control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the first reference voltage while the control signal is set at the first control signal and the second switch circuit is in the first connection state, and set the control signal at the first control signal and bring the second switch circuit into the first connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal, and during a second period, set the control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the third reference voltage while the control signal is set at the first control signal and the second switch circuit is in the second connection state, and set the control signal at the first control signal and bring the second switch circuit into the second connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal. The temperature measuring circuit is also provided with the following: a counter circuit configured to count a first frequency of the control signal during the first period, and a second frequency of the control signal during the second period; and a temperature obtaining unit configured to obtain the temperature based on the first and second frequencies.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1**A is a circuit diagram illustrating a temperature measuring circuit in State **1** according to a first embodiment;
[0013] FIG. **1**B is a circuit diagram illustrating the temperature measuring circuit in State **2** according to the first embodiment;
[0014] FIG. **1**C is a circuit diagram illustrating the temperature measuring circuit in State **3** according to the first embodiment;
[0015] FIG. **1**D is a circuit diagram illustrating the temperature measuring circuit in State **4** according to the first embodiment;
[0016] FIG. **2**A is a flowchart showing the operation of the temperature measuring circuit according to the first embodiment;
[0017] FIG. **2**B is a flowchart showing the operation of the temperature measuring circuit according to the first embodiment;
[0018] FIG. **3**A is a timing chart showing the following time variations during a first period T**1**: time variations of a first input voltage Vin**1** and a second input voltage Vin**2**, which are input to a comparator provided in the temperature measuring circuit according to the first embodiment; time variations of an inter-terminal voltage Vc, which is generated by a capacitor provided in the temperature measuring circuit; and time variations of a control signal S**1**, which is output by a control circuit provided in the temperature measuring circuit;

[0019] FIG. **3**B is a timing chart showing the following time variations during a second period T**2**: time variations of the first input voltage Vin**1** and the second input voltage Vin**2**, which are input to the comparator provided in the temperature measuring circuit according to the first embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; and time variations of the control signal S**1**, which is output by the control circuit provided in the temperature measuring circuit;
[0020] FIG. **4** is a circuit diagram of a third voltage supply provided in the temperature measuring circuit according to the first embodiment;
[0021] FIG. **5**A is a circuit diagram illustrating a temperature measuring circuit in State **1***a* according to a second embodiment;
[0022] FIG. **5**B is a circuit diagram illustrating the temperature measuring circuit in State **2***a* according to the second embodiment;
[0023] FIG. **5**C is a circuit diagram illustrating the temperature measuring circuit in State **1***b* according to the second embodiment;
[0024] FIG. **5**D is a circuit diagram illustrating the temperature measuring circuit in State **2***b* according to the second embodiment;
[0025] FIG. **5**E is a circuit diagram illustrating the temperature measuring circuit in State **3***a* according to the second embodiment;
[0026] FIG. **5**F is a circuit diagram illustrating the temperature measuring circuit in State **4***a* according to the second embodiment;
[0027] FIG. **5**G is a circuit diagram illustrating the temperature measuring circuit in State **3***b* according to the second embodiment;
[0028] FIG. **5**H is a circuit diagram illustrating the temperature measuring circuit in State **4***b* according to the second embodiment;
[0029] FIG. **6**A is a circuit diagram illustrating a comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being a non-inverting input terminal and an inverting input terminal, respectively;
[0030] FIG. **6**B is a circuit diagram illustrating the comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being an inverting input terminal and a non-inverting input terminal, respectively;
[0031] FIG. **7**A is a flowchart showing the operation of the temperature measuring circuit according to the second embodiment;
[0032] FIG. **7**B is a flowchart showing the operation of the temperature measuring circuit according to the second embodiment;
[0033] FIG. **8**A is a timing chart showing the following time variations during the first period T**1**: time variations of the first input voltage Vin**1** and the second input voltage Vin**2**, which are input to a comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by a capacitor provided in the temperature measuring circuit; time variations of the control signal S**1**, which is output by a control circuit provided in the temperature measuring circuit; and time variations of a control signal S**3**, which is output by a switching control circuit provided in the temperature measuring circuit; and
[0034] FIG. **8**B is a timing chart showing the following time variations during the second period T**2**: time variations of the first input voltage Vin**1** and the second input voltage Vin**2**, which are input to the comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; time variations of the control signal S**1**, which is output by the control circuit provided in the temperature measuring circuit; and time variations of the control signal S**3**, which is output by the switching control circuit provided in the temperature measuring circuit.
DETAILED DESCRIPTION OF THE INVENTION

[0035] The embodiments of the present disclosure will be described with reference to the drawings. It is noted that identical or equivalent constituents will be denoted by the same signs throughout the drawings, and the descriptions of redundancies will be omitted.

1 First Embodiment

1.1 Temperature Measuring Circuit

[0036] FIG. **1**A is a circuit diagram illustrating a temperature measuring circuit in State **1** according to a first embodiment. FIG. **1**B is a circuit diagram illustrating the temperature measuring circuit in State **2** according to the first embodiment. FIG. **1**C is a circuit diagram illustrating the temperature measuring circuit in State **3** according to the first embodiment. FIG. **1**D is a circuit diagram illustrating the temperature measuring circuit in State **4** according to the first embodiment.

[0037] The temperature measuring circuit, **1**, according to the first embodiment illustrated in FIGS. **1**A to **1**D measures a temperature and outputs the measured temperature.

[0038] The temperature measuring circuit **1** is formed on a semiconductor substrate and is incorporated in a semiconductor integrated circuit. The semiconductor substrate is a complementary metal oxide semiconductor (CMOS) wafer for instance. The temperature measuring circuit **1** does not need to be formed on a semiconductor substrate and does not need to be incorporated in a semiconductor integrated circuit.

[0039] As illustrated in FIGS. **1**A to **1**D, the temperature measuring circuit **1** includes a capacitor **11**, a first current supply **21**, a second current supply **22**, a first voltage supply **31**, a second voltage supply **32**, a third voltage supply **33**, a comparator circuit **41**, a control circuit **42**, a counter circuit **43**, a memory **44**, a temperature obtaining unit **45**, a first switch circuit **51**, and a second switch circuit **52**.

[0040] The capacitor **11** includes terminals **11***a* and **11***b*. The first current supply **21** includes a negative electrode **21***a* and a positive electrode **21***b*. The second current supply **22** includes a negative electrode **22***a* and a positive electrode **22***b*. The first voltage supply **31** includes a negative electrode **31***a* and a positive electrode **31***b*. The second voltage supply **32** includes a negative electrode **32***a* and a positive electrode **32***b*. The third voltage supply **33** includes a negative electrode **33***a* and a positive electrode **33***b*. The comparator circuit **41** includes a first input terminal **41***a*, a second input terminal **41***b*, and an output terminal **41***c*. The control circuit **42** includes an input terminal **42***a* and an output terminal **42***b*. The first switch circuit **51** includes terminals **51***a*, **51***b*, **51***c*, **51***d*, **51***e*, **51***f*, **51***g*, **51***h*, and **51***i*. The second switch circuit **52** includes terminals **52***a*, **52***b*, and **52***c*.

[0041] The terminal **11***b* of the capacitor **11** is grounded. The terminal **11***a* of the capacitor **11** is electrically connected to the terminal **51***c* of the first switch circuit **51**.

[0042] The negative electrode **21***a* of the first current supply **21** is electrically connected to a power supply. The positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **51***a* of the first switch circuit **51**. The negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **51***b* of the first switch circuit **51**. The positive electrode **22***b* of the second current supply **22** is grounded. The negative electrode **31***a* of the first voltage supply **31** is grounded. The positive electrode **31***b* of the first voltage supply **31** is electrically connected to the terminal **52***a* of the second switch circuit **52**. The negative electrode **32***a* of the second voltage supply **32** is grounded. The positive electrode **32***b* of the second voltage supply **32** is electrically connected to the terminal **51***h* of the first switch circuit **51**. The negative electrode **33***a* of the third voltage supply **33** is grounded. The positive electrode **33***b* of the third voltage supply **33** is electrically connected to the terminal **52***b* of the second switch circuit **52**. The first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are electrically connected to the terminals **51***f* and **51***i* of the first switch circuit **51**, respectively. The output terminal **41***c* of the comparator circuit **41** is electrically connected to the input terminal **42***a* of the comparator circuit **42**. The output terminal **42***b* of the control circuit **42** is electrically connected to the counter circuit **43**, the first switch circuit **51**, and the second switch circuit **52**. The terminal **52***c* of the second switch circuit **52** is electrically connected to the terminal **51***d* of the first switch circuit **51**. The counter circuit **43** is electrically connected to the control circuit **42**. The memory **44** is electrically connected to the counter circuit **43**. The temperature obtaining unit **45** is electrically connected to the counter circuit **43**.

[0043] The capacitor **11** stores electric charge. The amount of stored electric charge increases by the amount of electric charge carried by a current flowing into the terminal **11***a* of the capacitor **11** and decreases by the amount of electric charge carried by a current flowing out of the terminal **11***a* of the capacitor **11**. The capacitor **11** generates an inter-terminal voltage Vc corresponding to the amount of stored electric charge between the terminal **11***b* and the terminal **11***a*.

[0044] The first current supply **21** feeds a first current I**1** flowing into the negative electrode **21***a* and flowing out of the positive electrode **21***b*.

[0045] The second current supply **22** feeds a second current I**2** flowing into the negative electrode **22***a* and flowing out of the positive electrode **22***b*.

[0046] The first current supply **21** and the second current supply **22** are temperature-proportional current supplies, temperature-insensitive current supplies, process-variations-insensitive current supply, or other things. The first current supply **21** and the second current supply **22** that are process-variations-insensitive current supplies enable a temperature to be measured with high accuracy even when there are process variations in the semiconductor substrate on which the temperature measuring circuit **1** is formed.

[0047] The first voltage supply **31** generates a first reference voltage Vref between the negative electrode **31***a* and the positive electrode **31***b*.

[0048] The second voltage supply **32** generates a second reference voltage Vref/M between the negative electrode **32***a* and the positive electrode **32***b*.

[0049] The third voltage supply **33** generates a third reference voltage Vbe between the negative electrode **33***a* and the positive electrode **33***b*.

[0050] The first reference voltage Vref and the second reference voltage Vref/M are stable with respect to temperature. The third reference voltage Vbe is a voltage corresponding to temperature. The second reference voltage Vref/M is lower than the first reference voltage Vref and is 1/M times of the first reference voltage Vref. M is a positive value greater than 1.

[0051] The comparator circuit **41** outputs, from the output terminal **41***c*, an output voltage Vout indicating the levels of a first input voltage Vin**1** input to the first input terminal **41***a*, and a second input voltage Vin**2** input to the second input terminal **41***b*. The first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are a non-inverting input terminal and an inverting input terminal, respectively. The comparator circuit **41** thus outputs a first output voltage High as the output voltage Vout when the first input voltage Vin**1** is higher than the second input voltage Vin**2**, and the comparator circuit **41** outputs a second output voltage Low as the output voltage Vout when the first input voltage Vin**1** is lower than the second input voltage Vin**2**.

[0052] The control circuit **42** receives the output voltage Vout, which is output from the output terminal **41***c* of the comparator circuit **41**, from the input terminal **42***a*. The control circuit **42** outputs the control signals S**1** and S**2** from the output terminal **42***b*. The control circuit **42** switches the outgoing control signal S**1** between a first control signal High and a second control signal Low in synchronization with changes of the received output voltage Vout from the first output voltage High to the second output voltage Low. The control circuit **42** switches the outgoing control signal S**2** between the first control signal High and the second control signal Low.

[0053] The counter circuit **43** receives the control signal S**1** output from the output terminal **42***b* of the control circuit **42**. The counter circuit **43** counts the frequency of the received control signal S**1** and outputs the counted frequency.

[0054] The temperature obtaining unit **45** receives the frequency output from the counter circuit **43**. The temperature obtaining unit **45** obtains a temperature from the received frequency.

[0055] The first switch circuit **51** receives the control signal S**1** output from the output terminal **42***b* of the control circuit **42**.

[0056] Upon receiving the control signal S**1** being the first control signal High, the first switch circuit **51** brings its terminal **51***a* into electrical continuity with its terminal **51***c* and does not bring its terminal **51***b* into electrical continuity with its terminal **51***c*, as illustrated in FIG. **1**A and FIG. **1**C. The first switch circuit **51** thus causes the first current I**1** flowing out of the positive electrode **21***b* of the first current supply **21** to flow into the terminal **11***a* of the capacitor **11**, thus charging the capacitor **11** through the first current I**1**. This raises the inter-terminal voltage Vc, which is generated by the capacitor **11**, along with time.

[0057] Upon receiving the control signal S**1** being the second control signal Low, the first switch circuit **51** brings its terminal **51***b* into electrical continuity with its terminal **51***c* and does not bring its terminal **51***a* into electrical continuity with its terminal **51***c*, as illustrated in FIGS. **1**B and **1**D. The first switch circuit **51** thus causes the second current I**2** flowing into the negative electrode **22***a* of the second current supply **22** to flow out of the terminal **11***a* of the capacitor **11**, thus discharging the capacitor **11** through the second current I**2**. This lowers the inter-terminal voltage Vc, which is generated by the capacitor **11**, along with time.

[0058] In addition, upon receiving the control signal S**1** being the first control signal High, the first switch circuit **51** brings its terminal **51***d* into electrical continuity with its terminal **51***f* and does not bring its terminal **51***e* into electrical continuity with its terminal **51***f*, as illustrated in FIGS. **1**A and **1**C. The first switch circuit **51** thus sets the first input voltage Vin**1** input to the first input terminal **41***a* of the comparator circuit **41**, at the reference voltage Vref or Vbe selected by the second switch circuit **52**.

[0059] Upon receiving the control signal S**1** being the second control signal Low, the first switch circuit **51** brings its terminal **51***e* into electrical continuity with its terminal **51***f* and does not bring its terminal **51***d* into electrical continuity with its terminal **51***f*, as illustrated in FIGS. **1**B and **1**D. The first switch circuit **51** thus sets the first input voltage Vin**1** at the inter-terminal voltage Vc generated by the capacitor **11**.

[0060] In addition, upon receiving the control signal S**1** being the first control signal High, the first switch circuit **51** brings its terminal **51***g* into electrical continuity with its terminal **51***i* and does not bring its terminal **51***h* into electrical continuity with its terminal **51***i*, as illustrated in FIGS. **1**A and **1**C. The first switch circuit **51** thus sets the second input voltage Vin**2** at the inter-terminal voltage Vc generated by the capacitor **11**.

[0061] Upon receiving the control signal S**1** being the second control signal Low, the first switch circuit **51** brings its terminal **51***h* into electrical continuity with its terminal **51***i* and does not bring its terminal **51***g* into electrical continuity with its terminal **51***i*, as illustrated in FIGS. **1**B and **1**D. The first switch circuit **51** thus sets the second input voltage Vin**2** at the second reference voltage Vref/M generated by the second voltage supply **32**.

[0062] As described above, upon receiving the control signal S**1** being the first control signal High, the first switch circuit **51** sets the two input voltages Vin**1** and Vin**2**, which are compared by the comparator circuit **41**, at the reference voltage Vref or Vbe, which is selected by the second switch circuit **52**, and the inter-terminal voltage Vc, which is generated by the comparator circuit **11**. Upon receiving the control signal S**1** being the second control signal Low, the first switch circuit **51** sets the two input voltages Vin**1** and Vin**2**, which are compared by the comparator circuit **41**, at the inter-terminal voltage Vc, which is generated by the capacitor **11**, and the second reference voltage Vref/M, which is generated by the second voltage supply **32**.

[0063] The second switch circuit **52** receives the control signal S**2** output from the output terminal **42***b* of the control circuit **42**.

[0064] Upon receiving the control signal S**2** being the first control signal High, the second switch circuit **52** brings its terminal **52***a* into electrical continuity with its terminal **52***c* and does not bring its terminal **52***b* into electrical continuity with its terminal **52***c*, as illustrated in FIG. **1**A, FIG. **1**B, and FIG. **1**D. The second switch circuit **52** thus selects the first reference voltage Vref generated by the first voltage supply **31**.

[0065] Upon receiving the control signal S**2** being the second control signal Low, the second switch circuit **52** brings its terminal **52***b* into electrical continuity with its terminal **52***c* and does not bring its terminal **52***a* into electrical continuity with its terminal **52***c*, as illustrated in FIG. **1**C. The second switch circuit **52** thus selects the third reference voltage Vref generated by the third voltage supply **33**.

[0066] As described above, the second switch circuit **52** switches between a first connection state in which a reference voltage selected by the second switch circuit **52** is the first reference voltage Vref generated by the first voltage supply **31**, and a second connection state in which the reference voltage selected by the second switch circuit **52** is the third reference voltage Vbe generated by the third voltage supply **33**.

1.2 Operation of Temperature Measuring Circuit

[0067] FIGS. **2**A and **2**B are flowcharts showing the operation of the temperature measuring circuit according to the first embodiment.

[0068] The temperature measuring circuit **1** executes Steps S**101** to S**114** shown in FIGS. **2**A and **2**B.

[0069] The temperature measuring circuit **1** executes Steps S**101** to S**106** during a first period T**1**, which has a first measurement time TF**1**, to turn alternately into State **1** illustrated in FIG. **1**A and State **2** illustrated in FIG. **1**B. The temperature measuring circuit **1** executes Steps S**108** to S**113** during a subsequent second period T**2**, which has a second measurement time TF**2**, to turn alternately into State **3** illustrated in FIG. **1**C and State **4** illustrated in FIG. **1**D.

[0070] The temperature measuring circuit **1** executes Steps S**101** to S**103** during a period TIA, which is included in the first period T**1**, to turn into State **1** illustrated in FIG. **1**A. The temperature measuring circuit **1** executes Steps S**104** to S**106** during a period T**1**B, which is included in the first period T**1**, to turn into State **2** illustrated in FIG. **1**B.

[0071] The temperature measuring circuit **1** executes Steps S**108** to S**110** during a period T**2**A, which is included in the second period T**2**, to turn into State **3** illustrated in FIG. **1**C. The temperature measuring circuit **1** executes Steps S**111** to S**113** during a period T**2**B, which is included in the second period T**2**, to turn into State **4** illustrated in FIG. **1**D.

Charge during Period T**1**A

[0072] In Step S**101**, as illustrated in FIG. **1**A, the control circuit **42** sets the control signal S**1** at the first control signal High and sets the control signal S**2** at the first control signal High.

[0073] Accordingly, the positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the positive electrode **31***b* of the first voltage supply **31**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**.

[0074] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the first reference voltage Vref and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0075] In Step S**102**, the control circuit **42** determines whether a time TM**1** elapsed from the start of the first period T**1** is shorter than the first measurement time TF**1**. If determining that the time TM**1** is shorter than the first measurement time TF**1**, the control circuit **42** executes Step S**103**. If determining that the time TM**1** is equal to or longer than the first measurement time TF**1**, the control circuit **42** executes Step S**107**.

[0076] In Step S**103**, the control circuit **42** determines whether the inter-terminal voltage Vc is lower than the first reference voltage Vref on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the first reference voltage Vref, the control circuit **42** executes Step S**101**. If determining that the inter-terminal voltage Vc is equal to or higher than the first reference voltage Vref, the control circuit **42** executes Step S**104**.

[0077] From Steps S**101** to S**103**, the control circuit **42** causes the first current supply **21** to continue charging the capacitor **11** and causes the comparator circuit **41** to continue comparing the first reference voltage Vref and the inter-terminal voltage Vc until the time TM**1** reaches the first measurement time TF**1**, or until the inter-terminal voltage Vc rises to the first reference voltage Vref. The control circuit **42** ends the first period T**1** and starts the second period T**2** in synchronization with that the time TM**1** reaches the first measurement time TF**1**. The control circuit **42** ends the period T**1**A and starts the period T**1**B in synchronization with that the inter-terminal voltage Vc rises to the first reference voltage Vref, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge during Period T**1**B

[0078] In Step S**104**, as illustrated in FIG. **1**B, the control circuit **42** sets the control signal S**1** at the second control signal Low and sets the control signal S**2** at the first control signal High. The control circuit **42** may set the control signal S**2** at the second control signal Low.

[0079] Accordingly, the negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**.

[0080] Accordingly, the second current supply **22** discharges the capacitor **11**. Further, the comparator circuit **41** compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0081] In subsequent Step S**105**, the control circuit **42** determines whether the time TM**1** is shorter than the first measurement time TF**1**. If determining that the time TM**1** is shorter than the first measurement time TF**1**, the control circuit **42** executes Step S**106**. If determining that the time TM**1** is equal to or longer than the first measurement time TF**1**, the control circuit **42** executes Step S**107**.

[0082] In Step S**106**, the control circuit **42** determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit **42** executes Step S**104**. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit **42** executes Step S**101**.

[0083] From Steps S**104** to S**106**, the control circuit **42** causes the second current supply **22** to continue discharging the capacitor **11** and causes the comparator circuit **41** to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM**1** reaches the first measurement time TF**1**, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit **42** ends the first period T**1** and starts the second period T**2** in synchronization with that the time TM**1** reaches the first measurement time TF**1**. The control circuit **42** ends the period T**1**B and starts the period T**1**A in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Counting During First Period T**1**

[0084] From Steps S**101** to S**106**, the control circuit **42** sets the control signal S**1** at the second control signal Low in response to the output voltage Vout indicating that the inter-terminal voltage Vc has risen to the first reference voltage Vref while the control signal S**1** is the first reference voltage Vref, and while the second switch circuit **52** is in the first connection state. The control circuit **42** sets the control signal S**1** at the first control signal High and brings the second switch circuit **52** into the first connection state in response to the output voltage Vout indicating that the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M while the control signal S**1** is the second control signal Low.

[0085] That the inter-terminal voltage Vc has risen to the first reference voltage Vref is indicated by a change of the output voltage Vout from the first output voltage High to the second output voltage Low. That the inter-terminal voltage Vc has lowered to the second output voltage Low is indicated by a change of the output voltage Vout from the first output voltage High to the second output voltage Low.

[0086] As such, the control signal S**1** changes between the first control signal High and the second control signal Low periodically during the first period T**1**. The counter circuit **43** counts how many times the control signal S**1** has changed between the first control signal High and the second control signal Low during the first period T**1** (hereinafter, referred to as "counts"), and the counter circuit **43** writes the counts into a first address of the memory **44**.

Reset of Counter Circuit and Change of Count Writing Destination

[0087] In Step S**107**, the counter circuit **43** is reset, and the destination into which the foregoing number of periodic changes counted by the counter circuit **43** is written is changed from the first address to a second address.

Charge During Period T**2**A

[0088] In Step S**108**, the control circuit **42** sets the control signal S**1** at the first control signal High and sets the control signal S**2** at the second control signal Low, as illustrated in FIG. **1**C.

[0089] Accordingly, the positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the positive electrode **33***b* of the third voltage supply **33**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**.

[0090] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the third reference voltage Vbe and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0091] In subsequent Step S**109**, the control circuit **42** determines whether a time TM**2** elapsed from the start of the second period T**2** is shorter than the second measurement time TF**2**. If determining that the time TM**2** is shorter than the second measurement time TF**2**, the control circuit **42** executes Step S**110**. If determining that the time TM**2** is equal to or longer than the second measurement time TF**2**, the control circuit **42** executes Step S**114**.

[0092] In Step S**110**, the control circuit **42** determines whether the inter-terminal voltage Vc is lower than the third reference voltage Vbe on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the third reference voltage Vbe, the control circuit **42** executes Step S**108**. If determining that the inter-terminal voltage Vc is equal to or higher than the third reference voltage Vbe, the control circuit **42** executes Step S**111**.

[0093] From Steps S**108** to S**110**, the control circuit **42** causes the first current supply **21** to continue charging the capacitor **11** and causes the comparator circuit **41** to continue comparing the third reference voltage Vbe and the inter-terminal voltage Vc until the time TM**2** reaches the second measurement time TF**2**, or until the inter-terminal voltage Vc rises to the third reference voltage Vbe. The control circuit **42** ends the second period T**2** in synchronization with that the time TM**2** reaches the second measurement time TF**2**. The control circuit **42** ends the period T**2**A and starts the period T**2**B in synchronization with that the inter-terminal voltage Vc rises to the third reference voltage Vbe, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T**2**B

[0094] In Step S**111**, as illustrated in FIG. **1**D, the control circuit **42** sets the control signal S**1** at the second control signal Low and sets the control signal S**2** at the first control signal High. The control circuit **42** may set the control signal S**2** at the second control signal Low.

[0095] Accordingly, the negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**.

[0096] Accordingly, the second current supply **22** discharges the capacitor **11**. Further, the comparator circuit **41** compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0097] In subsequent Step S**112**, the control circuit **42** determines whether the time TM**2** is shorter than the second measurement time TF**2**. If determining that the time TM**2** is shorter than the second measurement time TF**2**, the control circuit **42** executes Step S**113**. If determining that the time TM**2** is equal to or longer than the second measurement time TF**2**, the control circuit **42** executes Step S**114**.

[0098] In Step S**113**, the control circuit **42** determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit **42** executes Step S**111**. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit **42** executes Step S**108**.

[0099] From Steps S**111** to S**113**, the control circuit **42** causes the second current supply **22** to continue discharging the capacitor **11** and causes the comparator circuit **41** to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM**2** reaches the second measurement time TF**2**, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit **42** ends the second period T**2** in synchronization with that the time TM**2** reaches the second measurement time TF**2**. The control circuit **42** ends the period T**2**B and starts the period T**2**A in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Counting During Second Period T**2**

[0100] From Steps S**107** to S**113**, the control circuit **42** sets the control signal S**1** at the second control signal Low in response to the output voltage Vout indicating that the inter-terminal voltage Vc has risen to the third reference voltage Vbe while the control signal S**1** is the first control signal High, and while the second switch circuit **52** is in the first connection state. The control circuit **42** sets the control signal S**1** at the first control signal High and brings the second switch circuit **52** into the first connection state in response to the output voltage Vout indicating that the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M while the control signal S**1** is the second control signal Low.

[0101] That the inter-terminal voltage Vc has risen to the third reference voltage Vbe is indicated by a change of the output voltage Vout from the first output voltage High to the second output voltage Low. That the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M is indicated by a change of the output voltage Vout from the first output voltage High to the second output voltage Low.

[0102] As such, the control signal S**1** changes between the first control signal High and the second control signal Low periodically during the second period T**2**. The counter circuit **43** counts how many times the control signal S**1** has changed between the first control signal High and the second control signal Low during the second period T**2**, and the counter circuit **43** writes the counts into the second address of the memory **44**.

Counts Readout

[0103] In Step S**114**, the counter circuit **43** reads the counts from the first and second addresses, and from the counts read from the first and second addresses, the counter circuit **43** obtains a first frequency F**1** of the control signal S**1** during the first period T**1**, and a second frequency F**2** of the control signal S**1** during the second period T**2**. Further, the temperature obtaining unit **45** obtains the temperature on the basis of the first frequency F**1** and second frequency F**2**.

1.3 Time Variations of First Input Voltage, Second Input Voltage, Inter-Terminal Voltage, and

Control Signal

Time Variations During First Period T**1**

[0104] FIG. **3**A is a timing chart showing the following time variations during the first period T**1**: time variations of the first input voltage Vin**1** and second input voltage Vin**2**, which are input to the comparator provided in the temperature measuring circuit according to the first embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; and time variations of the control signal S**1**, which is output by the control circuit provided in the temperature measuring circuit.

[0105] As shown in FIG. **3**A, in the temperature measuring circuit **1** in Steps S**101** to S**106**, the periods T**1**A and T**1**B come alternately during the first period T**1**.

[0106] During the period T**1**A, the first input voltage Vin**1** and the second input voltage Vin**2** are at the first reference voltage Vref and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the first reference voltage Vref. In addition, the control signal S**1** is kept at the first control signal High.

[0107] During the period T**1**B, the first input voltage Vin**1** and the second input voltage Vin**2** are at the inter-terminal voltage Vc and the second reference voltage Vref/M, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the first reference voltage Vref to the second reference voltage Vref/M. In addition, the control signal S**1** is kept at the second control signal Low.

[0108] During the periods T**1**A and T**1**B, the input terminal to which the inter-terminal voltage Vc is input is switched between the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41**. This enables switching between a state in which the fact that the inter-terminal voltage Vc has risen to a reference voltage can be detected, and a state in which the fact that the inter-terminal voltage Vc has lowered to the reference voltage can be detected. Also, during the periods T**1**A and T**1**B, the reference voltage compared to the inter-terminal voltage Vc is switched between the first reference voltage Vref and the second reference voltage Vref/M. This enables switching between a state in which the fact that the inter-terminal voltage Vc has risen to the first reference voltage Vref can be detected, and a state in which the fact that the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M can be detected.

[0109] The time necessary for the inter-terminal voltage Vc to rise from the second reference voltage Vref/M to the first reference voltage Vref does not depend on temperature. The time necessary for the inter-terminal voltage Vc to lower from the first reference voltage Vref to the second reference voltage Vref/M does not depend on temperature. Hence, the first frequency F**1** is independent of temperature and can be used as a reference frequency.

Time Variations During Second Period T**2**

[0110] FIG. **3**B is a timing chart showing the following time variations during the second period T**2**: time variations of the first input voltage Vin**1** and second input voltage Vin**2**, which are input to the comparator provided in the temperature measuring circuit according to the first embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; and time variations of the control signal S**1**, which is output by the control circuit provided in the temperature measuring circuit.

[0111] As shown in FIG. **3**B, in the temperature measuring circuit **1** in Steps S**108** to S**113**, the periods T**2**A and T**2**B come alternately during the second period T**2**.

[0112] During the period T**2**A, the first input voltage Vin**1** and the second input voltage Vin**2** are at the third reference voltage Vbe and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the third reference voltage Vbe. In addition, the control signal S**1** is kept at the first control signal High.

[0113] During the period T**2**B, the first input voltage Vin**1** and the second input voltage Vin**2** are at

the inter-terminal voltage Vc and the second reference voltage Vref/M, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the third reference voltage Vbe to the second reference voltage Vref/M. In addition, the control signal S**1** is kept at the second control signal Low.
[0114] During the periods T**2**A and T**2**B, the input terminal to which the inter-terminal voltage Vc is input is switched between the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41**. This enables switching between a state in which the fact that the inter-terminal voltage Vc has risen to a reference voltage can be detected, and a state in which the fact that the inter-terminal voltage Vc has lowered to the reference voltage can be detected. Also, during the periods T**2**A and T**2**B, the reference voltage compared to the inter-terminal voltage Vc is switched between the third reference voltage Vbe and the second reference voltage Vref/M. This enables switching between a state in which the fact that the inter-terminal voltage Vc has risen to the third reference voltage Vbe can be detected, and a state in which the fact that the inter-terminal voltage Vc has lowered to the second reference voltage Vref/M can be detected.
[0115] The time necessary for the inter-terminal voltage Vc to rise from the second reference voltage Vref/M to the third reference voltage Vbe depends on temperature. The time necessary for the inter-terminal voltage Vc to lower from the third reference voltage Vbe to the second reference voltage Vref/M depends on temperature. Hence, the second frequency F**2** is dependent on temperature, and unlike the first frequency F**1**, the second frequency F**2** can be used for specifying a temperature.
1.4 First Reference Voltage, Second Reference Voltage, and Third Reference Voltage
[0116] The first voltage supply **31** includes a bandgap reference circuit. The bandgap reference circuit generates temperature-independent voltage. The first voltage supply **31** extracts the generated voltage from the bandgap reference circuit and uses the extracted voltage as the first reference voltage Vref.
[0117] The second voltage supply **32** includes a bandgap reference circuit and a resistive voltage-dividing circuit. The bandgap reference circuit generates temperature-independent voltage. The resistive voltage-dividing circuit divides the generated voltage. The second voltage supply **32** extracts the divided voltage from the resistive voltage-dividing circuit and uses the extracted voltage as the second reference voltage Vref/M.
[0118] FIG. **4** is a circuit diagram of the third voltage supply provided in the temperature measuring circuit according to the first embodiment.
[0119] As illustrated in FIG. **4**, the third voltage supply **33** includes a current supply **61** and a bipolar transistor **62**. The current supply **61** has a negative electrode **61***a* and a positive electrode **61***b*. The bipolar transistor **62** has a base **62***a*, a collector **62***b*, and an emitter **62***c*.
[0120] The negative electrode **61***a* of the current supply **61** is electrically connected to a power supply. The positive electrode **61***b* of the current supply **61** is electrically connected to the emitter **62***c* of the bipolar transistor **62**. The bipolar transistor **62** is in diode connection. Thus, the base **62***a* and collector **62***b* of the bipolar transistor **62** are electrically connected to each other. The base **62***a* and the collector **62***b* are grounded.
[0121] The current supply **61** feeds a current flowing into the negative electrode **61***a* and flowing out of the positive electrode **61***b*. The current to be fed may or may not be temperature-proportional current. However, a semiconductor integrated circuit that incorporates the temperature measuring circuit **1** includes, in most cases, a current supply that feeds a temperature-proportional current. Hence, when the current to be fed is temperature-proportional current, the semiconductor integrated circuit can be downsized.
[0122] The bipolar transistor **62** feeds a current flowing into the emitter **62***c* and flowing out of the collector **62***b*.
[0123] The third voltage supply **33** extracts a voltage generated between the base **62***a* and emitter **62***c* from the emitter **62***c* and uses the extracted voltage as the third reference voltage Vbe.

[0124] The temperature coefficient of the voltage generated between the base **62***a* and emitter **62***c* is significantly smaller than the temperature coefficient of temperature-proportional current. Hence, even when the current to be fed is temperature-proportional current, the voltage extracted from the emitter **62***c* has a temperature coefficient close to the former temperature coefficient.

1.5 Temperature Obtainment

[0125] A ratio (Vbe−Vref/M):(Vref−Vref/M) coincides with the ratio between the first frequency F**1** and second frequency F**2**, i.e., F**1**:F**2**, where (Vbe-Vref/M) denotes the difference between the third reference voltage Vbe and second reference voltage Vref/M, and where (Vref-Vref/M) denotes the difference between the first reference voltage Vref and second reference voltage Vref/M. Accordingly, the temperature obtaining unit **45** determines the third reference voltage Vbe from the first reference voltage Vref, the second reference voltage Vref/M, the first frequency F**1**, and the second frequency F**2** in accordance with Mathematical Expression (11).

[Numeral1]

[00001]
$$\mathrm{Vbe} = \frac{F1}{F2} \times \left(\mathrm{Vref} - \frac{\mathrm{Vref}}{M}\right) + \frac{\mathrm{Vref}}{M} \quad (11)$$

[0126] The temperature obtaining unit **45** also obtains a temperature Temp from the temperature property of the determined third reference voltage Vbe. For example, the temperature obtaining unit **45** obtains the temperature Temp in accordance with Mathematical Expression (12) using a Boltzmann constant K, a saturation current Is, and a collector current Ic.

[Numeral2]

[00002]
$$\mathrm{Temp} = \frac{\mathrm{Vbe}}{K} \text{ .Math. } \log\left[\frac{I_S + I_C}{I_S}\right] \quad (12)$$

1.6 Advantages of Temperature Measuring Circuit

[0127] In the temperature measuring circuit **1**, the inter-terminal voltage Vc is kept at the second reference voltage Vref/M, which is a positive voltage, or higher. Thus, the voltage at the terminal **11***a* of the capacitor **11** is not a positive voltage. This facilitates forming the temperature measuring circuit **1** onto the semiconductor substrate.

[0128] In the temperature measuring circuit **1**, the first current supply **21** and the second current supply **22** do not have to be temperature-proportional current supplies. This enables various current supplies to be used as the first current supply **21** and the second current supply **22**. This also enables the temperature measuring circuit **1** to measure the temperature with high accuracy without being affected by variations in temperature-proportional current.

2 Second Embodiment

[0129] The following describes points in which the second embodiment is different from the first embodiment. With regard to what will not be described, a configuration similar to the configuration applied to the first embodiment will be applied to the second embodiment as well.

2.1 Temperature Measuring Circuit

[0130] FIG. **5**A is a circuit diagram illustrating a temperature measuring circuit in State **1***b* according to the second embodiment. FIG. **5**B is a circuit diagram illustrating the temperature measuring circuit in State **2***a* according to the second embodiment. FIG. **5**C is a circuit diagram illustrating the temperature measuring circuit in State **1***b* according to the second embodiment. FIG. **5**D is a circuit diagram illustrating the temperature measuring circuit in State **2***b* according to the second embodiment. FIG. **5**E is a circuit diagram illustrating the temperature measuring circuit in State **3***a* according to the second embodiment. FIG. **5**F is a circuit diagram illustrating the temperature measuring circuit in State **4***a* according to the second embodiment. FIG. **5**G is a circuit diagram illustrating the temperature measuring circuit in State **3***b* according to the second embodiment. FIG. **5**H is a circuit diagram illustrating the temperature measuring circuit in State **4***b* according to the second embodiment.

[0131] The temperature measuring circuit, **2**, according to the second embodiment illustrated in FIGS. **5**A to **5**H includes a switching control circuit **53**.

[0132] The switching control circuit **53** receives the control signal S**1** output from the output terminal **42***b* of the control circuit **42**. The switching control circuit **53** divides the received control signal S**1** and outputs a control signal S**3**. The switching control circuit **53** outputs the first control signal High or the second control signal Low as the control signal S**3**. The switching control circuit **53** sets the frequency of a change in the outgoing control signal S**3** between the first control signal High and the second control signal Low, at 1/N of the frequency of a change in the received control signal S**1** between the first control signal High and the second control signal Low. Here, N is an integer equal to or greater than 2. The switching control circuit **53** changes the outgoing control signal S**3** from the first control signal High to the second control signal Low when the control signal S**1** received with the outgoing control signal S**3** being the first control signal High is set at the first control signal High by the setpoint N. The switching control circuit **53** changes the outgoing control signal S**3** from the second control signal Low to the first control signal High when the control signal S**1** received with the outgoing control signal S**3** being the second control signal Low is set at the first control signal High by the setpoint N. The switching control circuit **53** is formed from a flip-flop circuit or other circuits.
[0133] The comparator circuit **41** in the temperature measuring circuit **2** receives the control signal S**3** output from the switching control circuit **53**. The comparator circuit **41** sets its first input terminal **41***a* and second input terminal **41***b* as a non-inverting input terminal and an inverting input terminal, respectively, while the received control signal S**3** is the second control signal Low. The comparator circuit **41** sets its first input terminal **41***a* and second input terminal **41***b* as an inverting input terminal and a non-inverting input terminal, respectively, while the received control signal S**3** is the first control signal High. Thus, while the received control signal S**3** is the second control signal Low, the comparator circuit **41** sets the output voltage Vout at the first output voltage High in response to the first input voltage Vin**1** higher than the second input voltage Vin**2**, and the comparator circuit **41** sets as the output voltage Vout at the second output voltage Low in response to the first input voltage Vin**1** lower than the second input voltage Vin**2**. Further, while the received control signal S**3** is the first control signal High, the comparator circuit **41** sets the output voltage Vout at the second output voltage Low in response to the first input voltage Vin**1** higher than the second input voltage Vin**2**, and the comparator circuit **41** sets the output voltage Vout at the first output voltage High in response to the first input voltage Vin**1** lower than the second input voltage Vin**2**.
[0134] The first switch circuit **51** in the temperature measuring circuit **2** includes terminals **51***j* and **51***k*, as illustrated in FIGS. **5**A through **5**H.
[0135] The terminal **51***j* of the first switch circuit **51** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**. The terminal **51***k* of the first switch circuit **51** is electrically connected to the terminal **52***c* of the second switch circuit **52**.
[0136] The first switch circuit **51** receives the control signal S**3** output from the switching control circuit **53**.
[0137] Upon receiving the control signal S**1** being the first control signal High, and receiving the control signal S**3** being the second control signal Low, the first switch circuit **51** brings its terminal **51***d* into electrical continuity with its terminal **51***f* and does not bring its terminals **51***e* and **51***j* into electrical continuity with its terminal **51***f*, as illustrated in FIGS. **5**A and **5**E. The first switch circuit **51** thus sets the first input voltage Vin**1** at the reference voltage Vref or Vbe selected by the second switch circuit **52**.
[0138] Upon receiving the control signal S**1** being the second control signal Low, and receiving the control signal S**3** being the second control signal Low, the first switch circuit **51** brings its terminal **51***e* into electrical continuity with its terminal **51***f* and does not bring its terminals **51***d* and **51***j* into electrical continuity with its terminal **51***f*, as illustrated in FIGS. **5**B and **5**F. The first switch circuit **51** thus sets the first input voltage Vin**1** at the inter-terminal voltage Vc generated by the capacitor **11**.

[0139] Upon receiving the control signal S**1** being the first control signal High, and receiving the control signal S**3** being the first control signal High, the first switch circuit **51** brings its terminal **51***e* into electrical continuity with its terminal **51***f* and does not bring its terminals **51***d* and **51***j* into electrical continuity with its terminal **51***f*, as illustrated in FIGS. **5**C and **5**G. The first switch circuit **51** thus sets the first input voltage Vin**1** at the inter-terminal voltage Vc generated by the capacitor **11**.

[0140] Upon receiving the control signal S**1** being the second control signal Low, and receiving the control signal S**3** being the first control signal High, the first switch circuit **51** brings its terminal **51***j* into electrical continuity with its terminal **51***f* and does not bring its terminals **51***d* and **51***e* into electrical continuity with its terminal **51***f*, as illustrated in FIGS. **5**D and **5**H. The first switch circuit **51** thus sets the first input voltage Vin**1** at the second reference voltage Vref/M generated by the second voltage supply **32**.

[0141] Upon receiving the control signal S**1** being the first control signal High, and receiving the control signal S**3** being the second control signal Low, the first switch circuit **51** brings its terminal **51***g* into electrical continuity with its terminal **51***i* and does not bring its terminals **51***h* and **51***k* into electrical continuity with its terminal **51***i*, as illustrated in FIGS. **5**A and **5**E. The first switch circuit **51** thus sets the second input voltage Vin**2** at the inter-terminal voltage Vc generated by the capacitor **11**.

[0142] Upon receiving the control signal S**1** being the second control signal Low, and receiving the control signal S**3** being the second control signal Low, the first switch circuit **51** brings its terminal **51***h* into electrical continuity with its terminal **51***i* and does not bring its terminals **51***g* and **51***k* into electrical continuity with its terminal **51***i*, as illustrated in FIGS. **5**B and **5**F. The first switch circuit **51** thus sets the second input voltage Vin**2** at the second reference voltage Vref/M generated by the second voltage supply **32**.

[0143] Upon receiving the control signal S**1** being the first control signal High, and receiving the control signal S**3** being the first control signal High, the first switch circuit **51** brings its terminal **51***k* into electrical continuity with its terminal **51***i* and does not bring its terminals **51***g* and **51***h* into electrical continuity with its terminal **51***i*, as illustrated in FIGS. **5**C and **5**G. The first switch circuit **51** thus sets the second input voltage Vin**2** at the reference voltage Vref or Vbe selected by the second switch circuit **52**.

[0144] Upon receiving the control signal S**1** being the second control signal Low, and receiving the control signal S**3** being the first control signal High, the first switch circuit **51** brings its terminal **51***g* into electrical continuity with its terminal **51***i* and does not bring its terminals **51***h* and **51***k* into electrical continuity with its terminal **51***i*, as illustrated in FIGS. **5**D and **5**H. The first switch circuit **51** thus sets the second input voltage Vin**2** at the inter-terminal voltage Vc generated by the capacitor **11**.

[0145] As described above, upon receiving the control signal S**1** being the first control signal High, and receiving the control signal S**3** being the second control signal Low, the first switch circuit **51** sets the two input voltages Vin**1** and Vin**2**, which are compared by the comparator circuit **41**, at the reference voltage Vref or Vbe, which is selected by the second switch circuit **52**, and the inter-terminal voltage Vc, which is generated by the comparator circuit **11**. Upon receiving the control signal S**1** being the second control signal Low, and receiving the control signal S**3** being the second control signal Low, the first switch circuit **51** sets the two input voltages Vin**1** and Vin**2**, which are compared by the comparator circuit **41**, at the inter-terminal voltage Vc, which is generated by the capacitor **11**, and the second reference voltage Vref/M, which is generated by the second voltage supply **32**. Upon receiving the control signal S**1** being the first control signal High, and receiving the control signal S**3** being the first control signal High, the first switch circuit **51** sets the two input voltages Vin**1** and Vin**2**, which are compared by the comparator circuit **41**, at the inter-terminal voltage Vc, which is generated by the comparator circuit **11**, and the reference voltage Vref or Vbe, which is selected by the second switch circuit **52**. Upon receiving the control signal S**1** being the second control signal Low, and receiving the control signal S**3** being the first control signal High, the first switch circuit **51** sets the two input voltages Vin**1** and Vin**2**, which are compared by the comparator circuit **41**, at the second reference voltage Vref/M, which is generated by the capacitor **32**, and the inter-terminal voltage Vc, which is generated by the second voltage supply **11**.

2.2 Comparator Circuit

[0146] FIG. **6**A is a circuit diagram illustrating the comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being a non-inverting input terminal and an inverting input terminal, respectively. FIG. **6**B is a circuit diagram illustrating the comparator circuit, which is provided in the temperature measuring circuit according to the second embodiment, with its first and second input terminals being an inverting input terminal and a non-inverting input terminal, respectively.

[0147] As illustrated in FIGS. **6**A and **6**B, the comparator circuit **41** includes a differential input circuit **71** and an output circuit **72**. The differential input circuit **71** includes a mirror pair **81**, a differential pair **82**, a constant-current supply **83**, and a switching circuit **84**. The mirror pair **81** has active elements **81***p* and **81***m*. The differential pair **82** has active elements **82***p* and **82***m*. The active elements **81***p*, **81***m*, **82***p*, and **82***m* are metal oxide semiconductor field-effect transistors (MOSFETs). The active elements **81***p*, **81***m*, **82***p*, and **82***m* may be active elements other than MOSFETs.

[0148] The mirror pair **81** feeds a reference current through one of the active elements **81***p* and **81***m* and feeds a current that is a duplicate of the reference current through the other of the active elements **81***p* and **81***m*.

[0149] The differential pair **82** feeds, through the active element **82***p*, a current corresponding to the first input voltage Vin**1** input to the first input terminal **41***a* of the comparator circuit **41** and feeds, through the active element **82***m*, a current corresponding to the second input voltage Vin**2** input to the second input terminal **41***b* of the comparator circuit **41**. The current fed through the active element **82***p* includes the current fed through the active element **81***p*. The current fed through the active element **82***m* includes the current fed through the active element **81***m*.

[0150] The constant-current supply **83** feeds a constant current. The constant current fed by the constant-current supply **83** is a current that is a merge of the current fed through the active element **82***p* and the current fed through the active element **82***m*.

[0151] The switching circuit **84** receives the control signal S**3**. Upon receiving the control signal S**3** being the second control signal Low, the switching circuit **84** brings the active element **81***m* into diode connection to feed the reference current through the active element **81***m* and feeds the duplicate of the reference current through the active element **81***p*, as illustrated in FIG. **6**A. Upon receiving the control signal S**3** being the second control signal Low, the switching circuit **84** also connects the output circuit **72** electrically to the active element **82***p* to feed, through the active element **82***p*, a current that is a merge of the current flowing into the active element **81***p* and current flowing out of the output circuit **72**. Upon receiving the control signal S**3** being the first control signal High, the switching circuit **84** brings the active element **81***p* into diode connection to feed the reference current through the active element **81***p* and feeds the duplicate of the reference current through the active element **81***m*, as illustrated in FIG. **6**B. Upon receiving the control signal S**3** being the first control signal High, the switching circuit **84** also connects the output circuit **72** electrically to the active element **82***m* to feed, through the active element **82***m*, a current that is a merge of the current flowing into the active element **81***m* and the current flowing out of the output circuit **72**.

[0152] The output circuit **72** outputs the output voltage Vout corresponding to an outgoing current, from the output terminal **41***c* of the comparator circuit **41**.

[0153] As described above, the switching circuit **84** sets the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** as a non-inverting input terminal and an inverting input terminal, respectively, while the control signal S**3** is the second control signal Low, and the switching circuit **84** sets the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** as an inverting input terminal and a non-inverting input terminal, respectively, while the received control signal S**3** is the first control signal High. Thus, while the control signal S**3** is the second control signal Low, the switching circuit **84** sets the output voltage Vout at the first output voltage High in response to the first input voltage Vin**1** higher than the second input voltage Vin**2**, and the switching circuit **84** sets the output voltage Vou at the second control signal Low in response to the first input voltage Vin**1** lower than the second input voltage Vin**2**. Further, while the control signal S**3** is the first control signal High, the switching circuit **84** sets the output voltage Vout at the second output voltage Low in response to the first input voltage Vin**1** higher than the second input voltage Vin**2**, and the switching circuit **84** sets the output voltage Vout at the first output voltage High in response to the first input voltage Vin**1** lower than the second input voltage Vin**2**.

2.3 Operation of Temperature Measuring Circuit

[0154] FIGS. **7**A and **7**B are flowcharts showing the operation of the temperature measuring circuit according to the second embodiment.

[0155] The temperature measuring circuit **2** executes Steps S**121** to S**154** shown in FIGS. **7**A and **7**B.

[0156] The temperature measuring circuit **2** executes Steps S**121** to S**136** during the first period T**1** to turn alternately into State **1***a* illustrated in FIG. **5**A and State **2***a* illustrated in FIG. **5**B, or turn alternately into State **1***b* illustrated in FIG. **5**C and State **2***b* illustrated FIG. **5**D. The temperature measuring circuit **2** executes Steps S**138** to S**153** during the subsequent second period T**2** to turn alternately into State **3***a* illustrated in FIG. **5**E and State **4***a* illustrated in FIG. **5**F, or turn alternately into State **3***b* illustrated in FIG. **5**G and State **4***b* illustrated FIG. **5**H.

[0157] The temperature measuring circuit **2** executes Steps S**122** to S**124** during the period T**1**A, which is included in the first period T**1**, to turn into State **1***a* illustrated in FIG. **5**A. The temperature measuring circuit **2** executes Steps S**125** to S**127** during the period T**1**B, which is included in the first period T**1**, to turn into State **2***a* illustrated in FIG. **5**B. The temperature measuring circuit **2** executes Steps S**130** to S**132** during a period TIC, which is included in the first period T**1**, to turn into State **1***b* illustrated in FIG. **5**C. The temperature measuring circuit **2** executes Steps S**133** to S**135** during a period T**1**D, which is included in the first period T**1**, to turn into State **2***b* illustrated in FIG. **5**D.

[0158] The temperature measuring circuit **2** executes Steps S**139** to S**141** during the period T**2**A, which is included in the second period T**2**, to turn into State **3***a* illustrated in FIG. **5**E. The temperature measuring circuit **2** executes Steps S**142** to S**144** during the period T**2**B, which is included in the second period T**2**, to turn into State **4***a* illustrated in FIG. **5**F. The temperature measuring circuit **2** executes Steps S**147** to S**149** during a period T**2**C, which is included in the second period T**2**, to turn into State **3***b* illustrated in FIG. **5**G. The temperature measuring circuit **2** executes Steps S**150** to S**152** during a period T**2**D, which is included in the second period T**2**, to turn into State **4***b* illustrated in FIG. **5**H.

Initialization of Measurement Number

[0159] In Step S**121**, the switching control circuit **53** initializes a measurement number i to zero. The measurement number i denotes how many times the control signal S**1** has been set at the first control signal High.

Charge During Period T**1**A

[0160] In subsequent Step S**122**, as illustrated in FIG. **5**A, the control circuit **42** sets the control signal S**1** at the first control signal High and sets the control signal S**2** at the first control signal High. In addition, the switching control circuit **53** sets the control signal S**3** at the second control signal Low. The switching control circuit **53** also increments the measurement number i.

[0161] Accordingly, the positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the positive electrode **31***b* of the first voltage supply **31**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **31***b* of the first voltage supply **31**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**.

[0162] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the first reference voltage Vref and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0163] In subsequent Step S**123**, the control circuit **42** determines whether the time TM**1** elapsed from the start of the first period T**1** is shorter than the first measurement time TF**1**. If determining that the time TM**1** is shorter than the first measurement time TF**1**, the control circuit **42** executes Step S**124**. If determining that the time TM**1** is equal to or longer than the first measurement time TF**1**, the control circuit **42** executes Step S**137**.

[0164] In Step S**124**, the control circuit **42** determines whether the inter-terminal voltage Vc is lower than the first reference voltage Vref on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the first reference voltage Vref, the control circuit **42** executes Step S**122**. If determining that the inter-terminal voltage Vc is equal to or higher than the first reference voltage Vref, the control circuit **42** executes Step S**125**.

[0165] From Steps S**122** to S**124**, the control circuit **42** causes the first current supply **21** to continue charging the capacitor **11** and causes the comparator circuit **41** to continue comparing the first reference voltage Vref and the inter-terminal voltage Vc until the time TM**1** reaches the first measurement time TF**1**, or until the inter-terminal voltage Vc rises to the first reference voltage Vref. The control circuit **42** ends the first period T**1** and starts the second period T**2** in synchronization with that the time TM**1** reaches the first measurement time TF**1**. The control circuit **42** ends the period T**1**A and starts the period T**1**B in synchronization with that the inter-terminal voltage Vc rises to the first reference voltage Vref, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T**1**B

[0166] In Step S**125**, as illustrated in FIG. **5**B, the control circuit **42** sets the control signal S**1** at the second control signal Low and sets the control signal S**2** at the first control signal High. The control circuit **42** may set the control signal S**2** at the second control signal Low. In addition, the switching control circuit **53** sets the control signal S**3** at the second control signal Low.

[0167] Accordingly, the negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**.

[0168] Accordingly, the second current supply **22** discharges the capacitor **11**. Further, the comparator circuit **41** compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0169] In subsequent Step S**126**, the control circuit **42** determines whether the time TM**1** is shorter than the first measurement time TF**1**. If determining that the time TM**1** is shorter than the first measurement time TF**1**, the control circuit **42** executes Step S**127**. If determining that the time TM**1** is equal to or longer than the first measurement time TF**1**, the control circuit **42** executes Step S**137**.

[0170] In Step S**127**, the control circuit **42** determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit **42** executes Step S**125**. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit **42** executes Step S**128**.

[0171] From Steps S**125** to S**127**, the control circuit **42** causes the second current supply **22** to continue discharging the capacitor **11** and causes the comparator circuit **41** to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM**1** reaches the first measurement time TF**1**, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit **42** ends the first period T**1** and starts the second period T**2** in synchronization with that the time TM**1** reaches the first measurement time TF**1**. The control circuit **42** ends the period T**1**B and starts the period T**1**A or TIC in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0172] In Step S**128**, the switching control circuit **53** determines whether the measurement number i has reached the setpoint N. If determining that the measurement number i has reached the setpoint N, the switching control circuit **53** executes Step S**129**. If determining that the measurement number i has not yet reached the setpoint N, the switching control circuit **53** executes Step S**122**.

[0173] From Steps S**121** to S**128**, until the measurement number i reaches the setpoint N, the control circuit **42** alternately performs the following two kinds of operations: one is causing the first current supply **21** to charge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the first reference voltage Vref, and the other is causing the second current supply **22** to discharge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the second reference voltage Vref/M. The control circuit **42** ends the period T**1**B and starts the period TIC in synchronization with that the measurement number i reaches the setpoint N.

Initialization of Measurement Number

[0174] In Step S**129**, the switching control circuit **53** initializes the measurement number i to zero.

Charge During Period TIC

[0175] In subsequent Step S**130**, as illustrated in FIG. **5**C, the control circuit **42** sets the control signal S**1** at the first control signal High and sets the control signal S**2** at the first control signal High. In addition, the switching control circuit **53** sets the control signal S**3** at the first control signal High. The switching control circuit **53** also increments the measurement number i.

[0176] Accordingly, the positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the positive electrode **31***b* of the first voltage supply **31**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, like in Step S**122**, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **31***b* of the first voltage supply **31**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**.

[0177] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the first reference voltage Vref and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0178] In subsequent Step S**131**, the control circuit **42** determines whether the time TM**1** elapsed from the start of the first period T**1** is shorter than the first measurement time TF**1**. If determining that the time TM**1** is shorter than the first measurement time TF**1**, the control circuit **42** executes Step S**132**. If determining that the time TM**1** is equal to or longer than the first measurement time TF**1**, the control circuit **42** executes Step S**137**.
[0179] In Step S**132**, the control circuit **42** determines whether the inter-terminal voltage Vc is lower than the first reference voltage Vref on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the first reference voltage Vref, the control circuit **42** executes Step S**130**. If determining that the inter-terminal voltage Vc is equal to or higher than the first reference voltage Vref, the control circuit **42** executes Step S**133**.
[0180] From Steps S**130** to S**132**, the control circuit **42** causes the first current supply **21** to continue charging the capacitor **11** and causes the comparator circuit **41** to continue comparing the first reference voltage Vref and the inter-terminal voltage Vc until the time TM**1** reaches the first measurement time TF**1**, or until the inter-terminal voltage Vc rises to the first reference voltage Vref. The control circuit **42** ends the first period T**1** and starts the second period T**2** in synchronization with that the time TM**1** reaches the first measurement time TF**1**. The control circuit **42** ends the period TIC and starts the period T**1**D in synchronization with that the inter-terminal voltage Vc rises to the first reference voltage Vref, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.
Discharge During Period T**1**D
[0181] In Step S**133**, the control circuit **42** sets the control signal S**1** at the second control signal Low and sets the control signal S**2** at the first control signal High. The control circuit **42** may set the control signal S**2** at the second control signal Low. In addition, the switching control circuit **53** sets the control signal S**3** at the first control signal High.
[0182] Accordingly, the negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **11***a* of the capacitor **11**, as illustrated in FIG. **5**D. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, like in Step S**125**, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**.
[0183] Accordingly, the second current supply **22** discharges the capacitor **11**. Further, the comparator circuit **41** compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.
[0184] In subsequent Step S**134**, the control circuit **42** determines whether the time TM**1** is shorter than the first measurement time TF**1**. If determining that the time TM**1** is shorter than the first measurement time TF**1**, the control circuit **42** executes Step S**135**. If determining that the time TM**1** is equal to or longer than the first measurement time TF**1**, the control circuit **42** executes Step S**137**.
[0185] In Step S**135**, the control circuit **42** determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit **42** executes Step S**133**. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit **42** executes Step S**136**.
[0186] From Steps S**133** to S**135**, the control circuit **42** causes the second current supply **22** to continue discharging the capacitor **11** and causes the comparator circuit **41** to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM**1** reaches

the first measurement time TF**1**, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit **42** ends the first period T**1** and starts the second period T**2** in synchronization with that the time TM**1** reaches the first measurement time TF**1**. The control circuit **42** ends the period T**1**D and starts the period TIC in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0187] In Step S**136**, the switching control circuit **53** determines whether the measurement number i has reached the setpoint N. If determining that the measurement number i has reached the setpoint N, the switching control circuit **53** executes Step S**121**. If determining that the measurement number i has not yet reached the setpoint N, the switching control circuit **53** executes Step S**130**.

[0188] From Steps S**129** to S**136**, until the measurement number i reaches the setpoint N, the control circuit **42** alternately performs the following two kinds of operations: one is causing the first current supply **21** to charge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the first reference voltage Vref, and the other is causing the second current supply **22** to discharge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the second reference voltage Vref/M. The control circuit **42** ends the period T**1**D and starts the period T**1**A in synchronization with that the measurement number i reaches the setpoint N.

[0189] From Steps S**121** to S**136**, during the first period T**1**, the non-inverting input terminal and inverting input terminal of the comparator circuit **41** are switched, and in synchronization with their switching, the voltages input to the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are switched. This can cancel the effect of an input offset resulting from variations in an active element. such as a transistor, provided in the comparator circuit **41**.

Reset of Counter Circuit and Change of Count Writing Destination

[0190] In Step S**137**, the counter circuit **43** is reset, and the destination into which the foregoing number of periodic changes counted by the counter circuit **43** is written is changed from the first address to the second address.

Initialization of Measurement Number

[0191] In Step S**138**, the switching control circuit **53** initializes the measurement number i to zero.

Charge During Period T**2**A

[0192] In Step S**139**, as illustrated in FIG. **5**E, the control circuit **42** sets the control signal S**1** at the first control signal High and sets the control signal S**2** at the second control signal Low. In addition, the switching control circuit **53** sets the control signal S**3** at the second control signal Low. The switching control circuit **53** also increments the measurement number i.

[0193] Accordingly, the positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the positive electrode **33***b* of the third voltage supply **33**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **33***b* of the third voltage supply **33**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**.

[0194] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the third reference voltage Vbe and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0195] In subsequent Step S**140**, the control circuit **42** determines whether the time TM**2** elapsed from the start of the second period T**2** is shorter than the second measurement time TF**2**. If determining that the time TM**2** is shorter than the second measurement time TF**2**, the control circuit **42** executes Step S**141**. If determining that the time TM**2** is equal to or longer than the second measurement time TF**2**, the control circuit **42** executes Step S**154**.

[0196] In Step S**141**, the control circuit **42** determines whether the inter-terminal voltage Vc is lower than the third reference voltage Vbe on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the third reference voltage Vbe, the control circuit **42** executes Step S**139**. If determining that the inter-terminal voltage Vc is equal to or higher than the third reference voltage Vbe, the control circuit **42** executes Step S**142**.

[0197] From Steps S**139** to S**141**, the control circuit **42** causes the first current supply **21** to continue charging the capacitor **11** and causes the comparator circuit **41** to continue comparing the third reference voltage Vbe and the inter-terminal voltage Vc until the time TM**2** reaches the second measurement time TF**2**, or until the inter-terminal voltage Vc rises to the third reference voltage Vbe. The control circuit **42** ends the second period T**2** in synchronization with that the time TM**2** reaches the second measurement time TF**2**. The control circuit **42** ends the period T**2**A and starts the period T**2**B in synchronization with that the inter-terminal voltage Vc rises to the third reference voltage Vbe, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T**2**B

[0198] In Step S**142**, the control circuit **42** sets the control signal S**1** at the second control signal Low and sets the control signal S**2** at the second control signal Low. The control circuit **42** may set the control signal S**2** at the first control signal High. In addition, the switching control circuit **53** sets the control signal S**3** at the second control signal Low.

[0199] Accordingly, the negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **11***a* of the capacitor **11**, as illustrated in FIG. **5**F Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as a non-inverting input terminal and an inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**.

[0200] Accordingly, the second current supply **22** discharges the capacitor **11**. Further, the comparator circuit **41** compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0201] In subsequent Step S**143**, the control circuit **42** determines whether the time TM**2** is shorter than the second measurement time TF**2**. If determining that the time TM**2** is shorter than the second measurement time TF**2**, the control circuit **42** executes Step S**144**. If determining that the time TM**2** is equal to or longer than the second measurement time TF**2**, the control circuit **42** executes Step S**154**.

[0202] In Step S**144**, the control circuit **42** determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit **42** executes Step S**142**. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit **42** executes Step S**145**.

[0203] From Steps S**142** to S**144**, the control circuit **42** causes the second current supply **22** to continue discharging the capacitor **11** and causes the comparator circuit **41** to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM**2** reaches the second measurement time TF**2**, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit **42** ends the second period T**2** in synchronization with that the time TM**2** reaches the second measurement time TF**2**. The control circuit **42** ends the period T**2**B and starts the period T**2**A or T**2**C in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0204] In Step S**145**, the switching control circuit **53** determines whether the measurement number i has reached the setpoint N. If determining that the measurement number i has reached the setpoint N, the switching control circuit **53** executes Step S**146**. If determining that the measurement number i has not yet reached the setpoint N, the switching control circuit **53** executes Step S**139**.

[0205] From Steps S**138** to S**145**, until the measurement number i reaches the setpoint N, the control circuit **42** alternately performs the following two kinds of operations: one is causing the first current supply **21** to charge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the third reference voltage Vbe; and the other is causing the second current supply **22** to discharge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the second reference voltage Vref/M. The control circuit **42** ends the period T**2**B and starts the period T**2**C in synchronization with that the measurement number i reaches the setpoint N.

Initialization of Measurement Number

[0206] In Step S**146**, the switching control circuit **53** initializes the measurement number i to zero.

Charge During Period T**2**C

[0207] In Step S**147**, as illustrated in FIG. **5**G, the control circuit **42** sets the control signal S**1** at the first control signal High and sets the control signal S**2** at the second control signal Low. In addition, the switching control circuit **53** sets the control signal S**3** at the first control signal High. The switching control circuit **53** also increments the measurement number i.

[0208] Accordingly, the positive electrode **21***b* of the first current supply **21** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the positive electrode **33***b* of the second voltage supply **33**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **33***b* of the third voltage supply **33**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**.

[0209] Accordingly, the first current supply **21** charges the capacitor **11**. Further, the comparator circuit **41** compares the third reference voltage Vbe and the inter-terminal voltage Vc and sets the output voltage Vout at the first output voltage High.

[0210] In subsequent Step S**148**, the control circuit **42** determines whether the time TM**2** elapsed from the start of the second period T**2** is shorter than the second measurement time TF**2**. If determining that the time TM**2** is shorter than the second measurement time TF**2**, the control circuit **42** executes Step S**149**. If determining that the time TM**2** is equal to or longer than the second measurement time TF**2**, the control circuit **42** executes Step S**154**.

[0211] In Step S**149**, the control circuit **42** determines whether the inter-terminal voltage Vc is lower than the third reference voltage Vbe on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is lower than the third reference voltage Vbe, the control circuit **42** executes Step S**147**. If determining that the inter-terminal voltage Vc is equal to or higher than the third reference voltage Vbe, the control circuit **42** executes Step S**150**.

[0212] From Steps S**147** to S**149**, the control circuit **42** causes the first current supply **21** to continue charging the capacitor **11** and causes the comparator circuit **41** to continue comparing the third reference voltage Vbe and the inter-terminal voltage Vc until the time TM**2** reaches the second measurement time TF**2**, or until the inter-terminal voltage Vc rises to the third reference voltage Vbe. The control circuit **42** ends the second period T**2** in synchronization with that the time TM**2** reaches the second measurement time TF**2**. The control circuit **42** ends the period T**2**C and starts the period T**2**D in synchronization with that the inter-terminal voltage Vc rises to the third reference voltage Vbe, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Discharge During Period T**2**D

[0213] In Step S**150**, as illustrated in FIG. **5**H, the control circuit **42** sets the control signal S**1** at the second control signal Low and sets the control signal S**2** at the second control signal Low. The control circuit **42** may set the control signal S**2** at the first control signal High. In addition, the switching control circuit **53** sets the control signal S**3** at the first control signal High.

[0214] Accordingly, the negative electrode **22***a* of the second current supply **22** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**. Further, the second input terminal **41***b* of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. Further, the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are set as an inverting input terminal and a non-inverting input terminal, respectively. Accordingly, the non-inverting input terminal of the comparator circuit **41** is electrically connected to the terminal **11***a* of the capacitor **11**. In addition, the inverting input terminal of the comparator circuit **41** is electrically connected to the positive electrode **32***b* of the second voltage supply **32**.

[0215] Accordingly, the second current supply **22** discharges the capacitor **11**. Further, the comparator circuit **41** compares the inter-terminal voltage Vc and the second reference voltage Vref/M and sets the output voltage Vout at the first output voltage High.

[0216] In subsequent Step S**151**, the control circuit **42** determines whether the time TM**2** is shorter than the second measurement time TF**2**. If determining that the time TM**2** is shorter than the second measurement time TF**2**, the control circuit **42** executes Step S**152**. If determining that the time TM**2** is equal to or longer than the second measurement time TF**2**, the control circuit **42** executes Step S**154**.

[0217] In Step S**152**, the control circuit **42** determines whether the inter-terminal voltage Vc is higher than the second reference voltage Vref/M on the basis of the output voltage Vout. If determining that the inter-terminal voltage Vc is higher than the second reference voltage Vref/M, the control circuit **42** executes Step S**150**. If determining that the inter-terminal voltage Vc is equal to or lower than the second reference voltage Vref/M, the control circuit **42** executes Step S**153**.

[0218] From Steps S**150** to S**152**, the control circuit **42** causes the second current supply **22** to continue discharging the capacitor **11** and causes the comparator circuit **41** to continue comparing the inter-terminal voltage Vc and the second reference voltage Vref/M until the time TM**2** reaches the second measurement time TF**2**, or until the inter-terminal voltage Vc lowers to the second reference voltage Vref/M. The control circuit **42** ends the second period T**2** in synchronization with that the time TM**2** reaches the second measurement time TF**2**. The control circuit **42** ends the period T**2**D and starts the period T**2**A or T**2**C in synchronization with that the inter-terminal voltage Vc lowers to the second reference voltage Vref/M, and that the output voltage Vout changes from the first output voltage High to the second output voltage Low.

Determination of Measurement Number

[0219] In Step S**153**, the switching control circuit **53** determines whether the measurement number i has reached the setpoint N. If determining that the measurement number i has reached the setpoint N, the switching control circuit **53** executes Step S**138**. If determining that the measurement number i has not yet reached the setpoint N, the switching control circuit **53** executes Step S**147**.

[0220] From Steps S**146** to S**153**, until the measurement number i reaches the setpoint N, the control circuit **42** alternately preforms the following two kinds of operations: one is causing the first current supply **21** to charge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the third reference voltage Vbe; and the other is causing the second current supply **22** to discharge the capacitor **11**, and causing the comparator circuit **41** to compare the inter-terminal voltage Vc and the second reference voltage Vref/M.

Counts Readout

[0221] In Step S**154**, the counter circuit **43** reads the counts from the first and second addresses, and from the counts read from the first and second addresses, the counter circuit **43** obtains the first frequency F**1** of the control signal S**1** during the first period T**1**, and the second frequency F**2** of the control signal S**1** during the second period T**2**. Further, the temperature obtaining unit **45** obtains the temperature on the basis of the first frequency F**1** and second frequency F**2**.

2.4 Time Variations of First Input Voltage, Second Input Voltage, Inter-Terminal Voltage, and Control Signal

Time Variations During First Period T**1**

[0222] FIG. **8**A is a timing chart showing the following time variations during the first period T**1**: time variations of the first input voltage Vin**1** and the second input voltage Vin**2**, which are input to the comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; time variations of the control signal S**1**, which is output by the control circuit provided in the temperature measuring circuit; and time variations of the control signal S**3**, which is output by the switching control circuit provided in the temperature measuring circuit.

[0223] As shown in FIG. **8**A, in the temperature measuring circuit **2** in Steps S**121** to S**136**, the periods T**1**A and T**1**B come alternately while the control signal S**3** during the first period T**1** is at the second control signal Low, i.e., while the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are a non-inverting input terminal and an inverting input terminal, respectively; moreover, the period TIC and T**1**D come alternately while the control signal S**3** during the first period T**1** is at the first control signal High, i.e., while the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are an inverting input terminal and a non-inverting input terminal, respectively.

[0224] During the period T**1**A, the first input voltage Vin**1** and the second input voltage Vin**2** are at the first reference voltage Vref and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the first reference voltage Vref. In addition, the control signal S**1** is kept at the first control signal High.

[0225] During the period T**1**B, the first input voltage Vin**1** and the second input voltage Vin**2** are at the inter-terminal voltage Vc and the second reference voltage Vref/M, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the first reference voltage Vref to the second reference voltage Vref/M. In addition, the control signal S**1** is kept at the second control signal Low.

[0226] During the period TIC, conversely to the period T**1**A, the first input voltage Vin**1** and the second input voltage Vin**2** are at the inter-terminal voltage Vc and the first reference voltage Vref, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the first reference voltage Vref. In addition, the control signal S**1** is kept at the first control signal High.

[0227] During the period T**1**D, conversely to the period T**1**B, the first input voltage Vin**1** and the second input voltage Vin**2** are at the second reference voltage Vref/M and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the first reference voltage Vref to the second reference voltage Vref/M. In addition, the control signal S**1** is kept at the second control signal Low.

[0228] During the periods T**1**A and TIC, the non-inverting input terminal and inverting input terminal of the comparator circuit **41** are switched. During the periods T**1**B and T**1**D, the non-inverting input terminal and inverting input terminal of the comparator circuit **41** are switched. This can cancel the effect of an input offset resulting from variations in an active element, such as a transistor, provided in the comparator circuit **41**.

Time Variations During Second Period T**2**

[0229] FIG. **8**B is a timing chart showing the following time variations during the second period T**2**: time variations of the first input voltage Vin**1** and the second input voltage Vin**2**, which are input to the comparator provided in the temperature measuring circuit according to the second embodiment; time variations of the inter-terminal voltage Vc, which is generated by the capacitor provided in the temperature measuring circuit; time variations of the control signal S**1**, which is output by the control circuit provided in the temperature measuring circuit; and time variations of the control signal S**3**, which is output by the switching control circuit provided in the temperature measuring circuit.

[0230] As shown in FIG. **8**B, in the temperature measuring circuit **2** in Steps S**138** to S**153**, the periods T**2**A and T**2**B come alternately while the control signal S**3** during the second period T**2** is at the second control signal Low, i.e., while the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are a non-inverting input terminal and an inverting input terminal, respectively; moreover, the period T**2**C and T**2**D come alternately while the control signal S**3** during the second period T**2** is at the first control signal High, i.e., while the first input terminal **41***a* and second input terminal **41***b* of the comparator circuit **41** are an inverting input terminal and a non-inverting input terminal, respectively.

[0231] During the period T**2**A, the first input voltage Vin**1** and the second input voltage Vin**2** are at the third reference voltage Vbe and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the third reference voltage Vbe. In addition, the control signal S**1** is kept at the first control signal High.

[0232] During the period T**2**B, the first input voltage Vin**1** and the second input voltage Vin**2** are at the inter-terminal voltage Vc and the second reference voltage Vref/M, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the third reference voltage Vbe to the second reference voltage Vref/M. In addition, the control signal S**1** is kept at the second control signal Low.

[0233] During the period T**2**C, conversely to the period T**2**A, the first input voltage Vin**1** and the second input voltage Vin**2** are at the inter-terminal voltage Vc and the third reference voltage Vbe, respectively. In addition, the inter-terminal voltage Vc rises along with time. Accordingly, the inter-terminal voltage Vc rises from the second reference voltage Vref/M to the third reference voltage Vbe. In addition, the control signal S**1** is kept at the first control signal High.

[0234] During the period T**2**D, conversely to the period T**2**B, the first input voltage Vin**1** and the second input voltage Vin**2** are at the second reference voltage Vref/M and the inter-terminal voltage Vc, respectively. In addition, the inter-terminal voltage Vc lowers along with time. Accordingly, the inter-terminal voltage Vc lowers from the third reference voltage Vbe to the second reference voltage Vref/M. In addition, the control signal S**1** is kept at the second control signal Low.

[0235] During the periods T**2**A and T**2**C, the non-inverting input terminal and inverting input terminal of the comparator circuit **41** are switched. During the periods T**2**B and T**2**D, the non-inverting input terminal and inverting input terminal of the comparator circuit **41** are switched. This can cancel the effect of an input offset resulting from variations in an active element, such as a transistor, provided in the comparator circuit **41**.

[0236] The present disclosure is not limited to the above-described embodiment. The present disclosure may be replaced with a configuration substantially identical to those described in the above-described embodiment, a configuration that provides the same action and effect as those described in the above-described embodiment, or a configuration that can achieve the same object as those described in the above-described embodiment.

[0237] While there have been described what are at present considered to be certain embodiments of the invention, it will be understood that various modifications may be made thereto, and it is intended that the appended claim cover all such modifications as fall within the true spirit and scope of the invention.

Claims

1. A temperature measuring circuit comprising: a capacitor configured to generate an inter-terminal voltage corresponding to an amount of stored electric charge; a first current supply configured to feed a first current; a second current supply configured to feed a second current; a first voltage supply configured to generate a first reference voltage; a second voltage supply configured to generate a second reference voltage; a third voltage supply configured to generate a third reference voltage corresponding to a temperature; a comparator circuit configured to output an output voltage indicating levels of two input voltages that are received; a first switch circuit configured to, upon receiving a first control signal, charge the capacitor through the first current and set the two input voltages at a selected reference voltage and the inter-terminal voltage, and upon receiving a second control signal, discharge the capacitor through the second current and set the two input voltages at the second reference voltage and the inter-terminal voltage; a second switch circuit configured to switch between a first connection state in which the selected reference voltage is set at the first reference voltage, and a second connection state in which the selected reference voltage is set at the third reference voltage; a control circuit configured to, during a first period, set a control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the first reference voltage while the control signal is set at the first control signal and the second switch circuit is in the first connection state, and set the control signal at the first control signal and bring the second switch circuit into the first connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal, and during a second period, set the control signal at the second control signal in response to the output voltage indicating that the inter-terminal voltage is higher than the third reference voltage while the control signal is set at the first control signal and the second switch circuit is in the second connection state, and set the control signal at the first control signal and bring the second switch circuit into the second connection state in response to the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage while the control signal is set at the second control signal; a counter circuit configured to count a first frequency of the control signal during the first period, and a second frequency of the control signal during the second period; and a temperature obtaining unit configured to obtain the temperature based on the first and second frequencies.

2. The temperature measuring circuit according to claim 1, wherein the first and second reference voltages are stable with respect to the temperature.

3. The temperature measuring circuit according to claim 1, wherein the second reference voltage is 1/M times of the first reference voltage, where M is a positive value greater than 1.

4. The temperature measuring circuit according to claim 1, wherein the third voltage supply includes a bipolar transistor having a base, a collector, and an emitter, and the third reference voltage is generated between the base and the emitter.

5. The temperature measuring circuit according to claim 1, wherein the temperature obtaining unit determines the third reference voltage from the first and second reference voltages and the first and second frequencies in accordance with following Mathematical Expression (1), and obtains the temperature from a temperature property of the third reference voltage:

[MathematicalExpression1]

$$Vbe = \frac{F1}{F2} \times (Vref - \frac{Vref}{M}) + \frac{Vref}{M}, \quad (1)$$

where Vref denotes the first reference voltage, where Vref/M denotes the second reference voltage, where F**1** denotes the first frequency, where F**2** denotes the second frequency, and where Vbe denotes the third reference voltage.

6. The temperature measuring circuit according to claim 1 comprising a switching control circuit configured to divide the control signal and output the control signal that has been divided.

7. The temperature measuring circuit according to claim 6, wherein the comparator circuit includes a first input terminal to which one of the two input voltages is input, a second input terminal to which another of the two input voltages is input, and a switching circuit configured to, while the control signal that has been divided is the first control signal, set the output voltage at a first output voltage in response to the one higher than the another, and set the output voltage at a second output voltage in response to the one lower than the another, and while the control signal that has been divided is the second control signal, set the output voltage at the second output voltage in response to the one higher than the another, and set the output voltage at the first output voltage in response to the one lower than the another.

8. The temperature measuring circuit according to claim 1, wherein the two input voltages are a first input voltage and a second input voltage, the comparator circuit sets the output voltage at a first output voltage in response to the first input voltage higher than the second input voltage, and sets the output voltage at the second output voltage in response to the first input voltage lower than the second input voltage, setting the selected reference voltage and the inter-terminal voltage at the two input voltages is equivalent to setting the selected reference voltage and the inter-terminal voltage at the first input voltage and the second input voltage, respectively, setting the second reference voltage and the inter-terminal voltage at the two input voltages is equivalent to setting the second reference voltage and the inter-terminal voltage at the second input voltage and the first input voltage, respectively, the output voltage indicating that the inter-terminal voltage is higher than the first reference voltage is equivalent to a change of the output voltage from the first reference voltage to the second output voltage, the output voltage indicating that the inter-terminal voltage is lower than the second reference voltage is equivalent to a change of the output voltage from the first output voltage to the second output voltage, and the output voltage indicating that the inter-terminal voltage is higher than the third reference voltage is equivalent to a change of the output voltage from the first output voltage to the second output voltage.